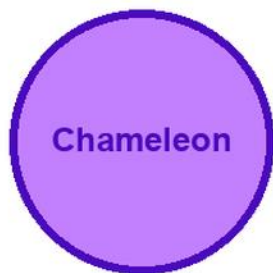


KITABU QUEST

Standard IV

SAYANSI



Cover scene

learners observing plants and simple experiments with a chameleon mascot

● Clear lessons

● Worked examples

● Fun activities

● Real-life problems

GRADE

IV

Title Page

KITABU QUEST Standard IV Sayansi

Tanzania TIE-BASIC learner book

Mascot: chameleon

This KITABU QUEST book turns the curriculum into short lessons, worked examples, guided activities, practice, review, and home links for Tanzania learners.

How To Use This Book

This book helps Standard IV learners study Sayansi one teachable topic at a time.

1. Start with the inquiry question and say what you already know.
2. Read the Learn section slowly and mark new words.
3. Try the worked example before reading the full solution.
4. Complete the activity and practice tasks.
5. Correct your answer using the success criteria.
6. Ask for support when your answer is still unclear.

Subject method: Chunguza, uliza maswali, jaribu kwa usalama, andika ushahidi, na eleza matokeo.

Learning Skills In This Book

Each topic builds knowledge, skill, values, communication, collaboration, critical thinking, creativity, digital awareness, and self-confidence.

Study actively: speak answers aloud, draw when useful, show your working, cite evidence, use local examples, and correct mistakes after feedback.

A strong answer is specific. It names the idea, gives evidence or method, applies it to the task, and checks whether it answers the question.

Table Of Contents

Use this table to move through the book one topic at a time. Each topic has a lesson opener, clear teaching, a model or example, guided work, practice, and checks for understanding.

1. Living Things Plants And Animals
2. Water Air And Weather
3. Food Health And Hygiene
4. Materials And Their Uses
5. Forces Movement And Simple Tools
6. Light Sound And Heat Around Us
7. Environment Care And Safety
8. Primary Science Project

As you finish each topic, return to the success criteria and correct one answer before moving on.

Living Things Plants And Animals: Lesson Opener

Learning focus: Living Things Plants And Animals

Country link: a safe Tanzania observation, model, data table, or investigation for Living Things Plants And Animals.

Progression focus: build concrete foundations using simple examples and teacher-guided correction; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain living things plants and animals using evidence

Inquiry questions:

- How can living things plants and animals help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Living Things Plants And Animals: Learn

Living Things Plants And Animals is learned by asking a clear question, observing carefully, recording evidence, and explaining what the evidence means.

Core idea:

- Start with the object, organism, material, body part, force, energy change, or environmental feature being studied.
- Use safe observations, labelled diagrams, simple measurements, or comparison tables.
- Keep evidence separate from guesses: evidence is what you saw, measured, read from a table, or labelled correctly.
- A conclusion should answer the question and mention one limit or next observation.
- Local examples from home, school, farms, water sources, weather, health, and the environment make the science useful.

Key words:

- Living Things Plants And Animals
- Living
- Things
- Plants
- Animals
- concept
- observation
- evidence

Living Things Plants And Animals: Worked Example

Investigation model for Living Things Plants And Animals: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Living Things Plants And Animals: Vocabulary And Study Skill

Use these words and study moves before you practise Living Things Plants And Animals. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Living Things Plants And Animals
- Living
- Things
- Plants
- Animals
- concept
- observation
- evidence

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Living Things Plants And Animals without a clear example, method, or evidence.

Living Things Plants And Animals: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Living Things Plants And Animals.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Living Things Plants And Animals: Practice And Correction

Practice for Living Things Plants And Animals:

Main outcome: investigate and explain living things plants and animals using evidence.

1. Write one clear science question for the topic.
2. Use a labelled diagram, observation table, or model to record evidence.
3. Write a conclusion that answers the question and uses topic vocabulary.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Living Things Plants And Animals: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Living Things Plants And Animals.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Living Things Plants And Animals: Outcome Check

Outcome check for Living Things Plants And Animals: Investigate and explain living things plants and animals using evidence.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Water Air And Weather: Lesson Opener

Learning focus: Water Air And Weather

Country link: a safe Tanzania observation, model, data table, or investigation for Water Air And Weather.

Progression focus: build concrete foundations using simple examples and teacher-guided correction; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain water air and weather using evidence

Inquiry questions:

- How can water air and weather help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Water Air And Weather: Learn

Water Air And Weather is learned by asking a clear question, observing carefully, recording evidence, and explaining what the evidence means.

Core idea:

- Start with the object, organism, material, body part, force, energy change, or environmental feature being studied.
- Use safe observations, labelled diagrams, simple measurements, or comparison tables.
- Keep evidence separate from guesses: evidence is what you saw, measured, read from a table, or labelled correctly.
- A conclusion should answer the question and mention one limit or next observation.
- Local examples from home, school, farms, water sources, weather, health, and the environment make the science useful.

Key words:

- Water Air And Weather
- Water
- Weather
- concept
- observation
- evidence
- safety
- conclusion

Water Air And Weather: Worked Example

Investigation model for Water Air And Weather: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Water Air And Weather: Vocabulary And Study Skill

Use these words and study moves before you practise Water Air And Weather. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Water Air And Weather
- Water
- Weather
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Water Air And Weather without a clear example, method, or evidence.

Water Air And Weather: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Water Air And Weather.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Water Air And Weather: Practice And Correction

Practice for Water Air And Weather:

Main outcome: investigate and explain water air and weather using evidence.

1. Write one clear science question for the topic.
2. Use a labelled diagram, observation table, or model to record evidence.
3. Write a conclusion that answers the question and uses topic vocabulary.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Water Air And Weather: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Water Air And Weather.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Water Air And Weather: Outcome Check

Outcome check for Water Air And Weather: Investigate and explain water air and weather using evidence.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Food Health And Hygiene: Lesson Opener

Learning focus: Food Health And Hygiene

Country link: a safe Tanzania observation, model, data table, or investigation for Food Health And Hygiene.

Progression focus: build concrete foundations using simple examples and teacher-guided correction; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain food health and hygiene using evidence

Inquiry questions:

- How can food health and hygiene help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Food Health And Hygiene: Learn

Food Health And Hygiene is learned by asking a clear question, observing carefully, recording evidence, and explaining what the evidence means.

Core idea:

- Start with the object, organism, material, body part, force, energy change, or environmental feature being studied.
- Use safe observations, labelled diagrams, simple measurements, or comparison tables.
- Keep evidence separate from guesses: evidence is what you saw, measured, read from a table, or labelled correctly.
- A conclusion should answer the question and mention one limit or next observation.
- Local examples from home, school, farms, water sources, weather, health, and the environment make the science useful.

Key words:

- Food Health And Hygiene
- Food
- Health
- Hygiene
- concept
- observation
- evidence
- safety

Food Health And Hygiene: Worked Example

Investigation model for Food Health And Hygiene: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Food Health And Hygiene: Vocabulary And Study Skill

Use these words and study moves before you practise Food Health And Hygiene. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Food Health And Hygiene
- Food
- Health
- Hygiene
- concept

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Food Health And Hygiene without a clear example, method, or evidence.

Food Health And Hygiene: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Food Health And Hygiene.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Food Health And Hygiene: Practice And Correction

Practice for Food Health And Hygiene:

Main outcome: investigate and explain food health and hygiene using evidence.

1. Write one clear science question for the topic.
2. Use a labelled diagram, observation table, or model to record evidence.
3. Write a conclusion that answers the question and uses topic vocabulary.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Food Health And Hygiene: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Food Health And Hygiene.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Food Health And Hygiene: Outcome Check

Outcome check for Food Health And Hygiene: Investigate and explain food health and hygiene using evidence.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Materials And Their Uses: Lesson Opener

Learning focus: Materials And Their Uses

Country link: a safe Tanzania observation, model, data table, or investigation for Materials And Their Uses.

Progression focus: build concrete foundations using simple examples and teacher-guided correction; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain materials and their uses using evidence

Inquiry questions:

- How can materials and their uses help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Materials And Their Uses: Learn

Materials And Their Uses is learned by asking a clear question, observing carefully, recording evidence, and explaining what the evidence means.

Core idea:

- Start with the object, organism, material, body part, force, energy change, or environmental feature being studied.
- Use safe observations, labelled diagrams, simple measurements, or comparison tables.
- Keep evidence separate from guesses: evidence is what you saw, measured, read from a table, or labelled correctly.
- A conclusion should answer the question and mention one limit or next observation.
- Local examples from home, school, farms, water sources, weather, health, and the environment make the science useful.

Key words:

- Materials And Their Uses
- Materials
- Their
- Uses
- concept
- observation
- evidence
- safety

Materials And Their Uses: Worked Example

Investigation model for Materials And Their Uses: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Materials And Their Uses: Vocabulary And Study Skill

Use these words and study moves before you practise Materials And Their Uses. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Materials And Their Uses
- Materials
- Their
- Uses
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Materials And Their Uses without a clear example, method, or evidence.

Materials And Their Uses: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Materials And Their Uses.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Materials And Their Uses: Practice And Correction

Practice for Materials And Their Uses:

Main outcome: investigate and explain materials and their uses using evidence.

1. Write one clear science question for the topic.
2. Use a labelled diagram, observation table, or model to record evidence.
3. Write a conclusion that answers the question and uses topic vocabulary.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Materials And Their Uses: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Materials And Their Uses.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Materials And Their Uses: Outcome Check

Outcome check for Materials And Their Uses: Investigate and explain materials and their uses using evidence.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Forces Movement And Simple Tools: Lesson Opener

Learning focus: Forces Movement And Simple Tools

Country link: a safe Tanzania observation, model, data table, or investigation for Forces Movement And Simple Tools.

Progression focus: build concrete foundations using simple examples and teacher-guided correction; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain forces movement and simple tools using evidence

Inquiry questions:

- How can forces movement and simple tools help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Forces Movement And Simple Tools: Learn

Forces Movement And Simple Tools explains how muscles help the body move and stay active.

Core idea:

- Muscles pull on bones to create movement at joints.
- Muscles usually work in pairs: one contracts while the other relaxes.
- Exercise, rest, clean food, and safe posture help muscles remain healthy.
- Sudden heavy work, poor warm-up, and unsafe movement can injure muscles.
- A labelled body diagram helps learners connect muscle action with daily movement.

Key words:

- Forces Movement And Simple Tools
- Forces
- Movement
- Simple
- Tools
- concept
- observation
- evidence

Forces Movement And Simple Tools: Worked Example

Model for Forces Movement And Simple Tools: Explain how the arm bends.

1. Hold one arm out and bend it slowly.
2. Feel the upper-arm muscle shorten as the elbow bends.
3. Straighten the arm and notice the movement changes.
4. Draw a simple arm diagram with bone, joint, and muscle labels.
5. Add one safety note about warm-up or avoiding sudden heavy lifting.

Conclusion: muscles pull on bones at joints to make movement.

Forces Movement And Simple Tools: Vocabulary And Study Skill

Use these words and study moves before you practise Forces Movement And Simple Tools. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Forces Movement And Simple Tools
- Forces
- Movement
- Simple
- Tools
- concept
- observation
- evidence

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Forces Movement And Simple Tools without a clear example, method, or evidence.

Forces Movement And Simple Tools: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Forces Movement And Simple Tools.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Forces Movement And Simple Tools: Practice And Correction

Practice for Forces Movement And Simple Tools:

Main outcome: investigate and explain forces movement and simple tools using evidence.

1. Draw and label a simple joint and muscle action.
2. Explain how a muscle helps movement by pulling.
3. List two habits that protect muscles during work or exercise.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Forces Movement And Simple Tools: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Forces Movement And Simple Tools.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Forces Movement And Simple Tools: Outcome Check

Outcome check for Forces Movement And Simple Tools: Investigate and explain forces movement and simple tools using evidence.

1. State how muscles create movement.
2. Name one joint or body movement example.
3. Write one habit that keeps muscles healthy.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Light Sound And Heat Around Us: Lesson Opener

Learning focus: Light Sound And Heat Around Us

Country link: a safe Tanzania observation, model, data table, or investigation for Light Sound And Heat Around Us.

Progression focus: build concrete foundations using simple examples and teacher-guided correction; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain light sound and heat around us using evidence

Inquiry questions:

- How can light sound and heat around us help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Light Sound And Heat Around Us: Learn

Light Sound And Heat Around Us explains thermal energy and temperature.

Core idea:

- Temperature tells how hot or cold something is.
- Heat moves from a hotter object to a cooler object.
- Thermometers must be read carefully and safely.
- Different scales, such as Celsius and Kelvin, can describe temperature.

Key words:

- Light Sound And Heat Around Us
- Light
- Sound
- Heat
- Around
- concept
- observation
- evidence

Light Sound And Heat Around Us: Worked Example

Investigation model for Light Sound And Heat Around Us: Read water temperature safely.

1. Put warm water in a beaker under teacher guidance.
2. Place a thermometer bulb in the water without touching the bottom.
3. Wait for the reading to stop changing.
4. Read the Celsius scale at eye level.

Conclusion: careful thermometer use gives a reliable temperature.

Light Sound And Heat Around Us: Vocabulary And Study Skill

Use these words and study moves before you practise Light Sound And Heat Around Us. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Light Sound And Heat Around Us
- Light
- Sound
- Heat
- Around
- concept
- observation
- evidence

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Light Sound And Heat Around Us without a clear example, method, or evidence.

Light Sound And Heat Around Us: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Light Sound And Heat Around Us.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Light Sound And Heat Around Us: Practice And Correction

Practice for Light Sound And Heat Around Us:

Main outcome: investigate and explain light sound and heat around us using evidence.

1. Read a thermometer diagram or classroom thermometer safely.
2. State the direction heat flows between hot and cold objects.
3. Compare heat and temperature in one sentence.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Light Sound And Heat Around Us: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Light Sound And Heat Around Us.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Light Sound And Heat Around Us: Outcome Check

Outcome check for Light Sound And Heat Around Us: Investigate and explain light sound and heat around us using evidence.

1. State the direction heat flows.
2. Describe safe thermometer reading.
3. Use the word temperature correctly.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Environment Care And Safety: Lesson Opener

Learning focus: Environment Care And Safety

Country link: a safe Tanzania observation, model, data table, or investigation for Environment Care And Safety.

Progression focus: build concrete foundations using simple examples and teacher-guided correction; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain environment care and safety using evidence

Inquiry questions:

- How can environment care and safety help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Environment Care And Safety: Learn

Environment Care And Safety begins with safe behaviour and accurate measurement.

Core idea:

- Physics uses measurements such as length, mass, time, temperature, and volume.
- A measurement is useful only when the number and unit are both written.
- Safety rules protect learners, equipment, and results.
- Read scales at eye level and record values immediately.

Local link: in Tanzania, careful measurement helps in health, building, farming, weather records, transport, and school laboratory work.

Key words:

- Environment Care And Safety
- Environment
- Care
- Safety
- concept
- observation
- evidence
- safety

Environment Care And Safety: Worked Example

Investigation model for Environment Care And Safety: Measure the length of a desk safely.

1. Choose a metre rule or tape measure.
2. Place zero at one end of the desk.
3. Read the scale at eye level.
4. Record the value with the unit, for example 120 cm.
5. Repeat once and compare readings.

Conclusion: a useful measurement includes a number, a unit, and a careful method.

Environment Care And Safety: Vocabulary And Study Skill

Use these words and study moves before you practise Environment Care And Safety. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Environment Care And Safety
- Environment
- Care
- Safety
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Environment Care And Safety without a clear example, method, or evidence.

Environment Care And Safety: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Environment Care And Safety.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Environment Care And Safety: Practice And Correction

Practice for Environment Care And Safety:

Main outcome: investigate and explain environment care and safety using evidence.

1. Name three laboratory safety rules.
2. Measure one classroom object and record the value with a unit.
3. Explain one error that can make a measurement unreliable.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Environment Care And Safety: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Environment Care And Safety.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Environment Care And Safety: Outcome Check

Outcome check for Environment Care And Safety: Investigate and explain environment care and safety using evidence.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Primary Science Project: Lesson Opener

Learning focus: Primary Science Project

Country link: a safe Tanzania observation, model, data table, or investigation for Primary Science Project.

Progression focus: build concrete foundations using simple examples and teacher-guided correction; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain primary science project using evidence

Inquiry questions:

- How can primary science project help you solve a real problem?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Primary Science Project: Learn

Primary Science Project is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Primary Science Project
- Primary
- Science
- Project
- concept
- observation
- evidence
- safety

Primary Science Project: Worked Example

Investigation model for Primary Science Project: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Primary Science Project: Vocabulary And Study Skill

Use these words and study moves before you practise Primary Science Project. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Primary Science Project
- Primary
- Science
- Project
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Tanzania.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Primary Science Project without a clear example, method, or evidence.

Primary Science Project: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Tanzania observation, model, data table, or investigation for Primary Science Project.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Primary Science Project: Practice And Correction

Practice for Primary Science Project:

Main outcome: investigate and explain primary science project using evidence.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Primary Science Project: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Primary Science Project.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Primary Science Project: Outcome Check

Outcome check for Primary Science Project: Investigate and explain primary science project using evidence.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Living Things Plants And Animals: Review Clinic 1

Use this review clinic to strengthen Sayansi without copying earlier answers.

1. Choose one idea from Living Things Plants And Animals that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Tanzania.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Water Air And Weather: Review Clinic 2

Use this review clinic to strengthen Sayansi without copying earlier answers.

1. Choose one idea from Water Air And Weather that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Tanzania.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Food Health And Hygiene: Review Clinic 3

Use this review clinic to strengthen Sayansi without copying earlier answers.

1. Choose one idea from Food Health And Hygiene that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Tanzania.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Materials And Their Uses: Review Clinic 4

Use this review clinic to strengthen Sayansi without copying earlier answers.

1. Choose one idea from Materials And Their Uses that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Tanzania.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Living Things Plants And Animals: Application Workshop 1

Application workshop 1 for Living Things Plants And Animals.

Grade move: build concrete foundations using simple examples and teacher-guided correction; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Water Air And Weather: Application Workshop 2

Application workshop 2 for Water Air And Weather.

Grade move: build concrete foundations using simple examples and teacher-guided correction; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Food Health And Hygiene: Application Workshop 3

Application workshop 3 for Food Health And Hygiene.

Grade move: build concrete foundations using simple examples and teacher-guided correction; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Materials And Their Uses: Application Workshop 4

Application workshop 4 for Materials And Their Uses.

Grade move: build concrete foundations using simple examples and teacher-guided correction; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Forces Movement And Simple Tools: Application Workshop 5

Application workshop 5 for Forces Movement And Simple Tools.

Grade move: build concrete foundations using simple examples and teacher-guided correction; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Light Sound And Heat Around Us: Application Workshop 6

Application workshop 6 for Light Sound And Heat Around Us.

Grade move: build concrete foundations using simple examples and teacher-guided correction; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Environment Care And Safety: Application Workshop 7

Application workshop 7 for Environment Care And Safety.

Grade move: build concrete foundations using simple examples and teacher-guided correction; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Primary Science Project: Application Workshop 8

Application workshop 8 for Primary Science Project.

Grade move: build concrete foundations using simple examples and teacher-guided correction; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Living Things Plants And Animals: Application Workshop 9

Application workshop 9 for Living Things Plants And Animals.

Grade move: build concrete foundations using simple examples and teacher-guided correction; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Water Air And Weather: Application Workshop 10

Application workshop 10 for Water Air And Weather.

Grade move: build concrete foundations using simple examples and teacher-guided correction; connect evidence, diagrams, safety, and conclusions.

Use this page to deepen the science idea with evidence.

1. Write one question that can be answered by observation, model, measurement, classification, or data reading.
2. Name safe materials, diagrams, samples, tables, or apparatus that would help.
3. Predict what you expect and explain why.
4. Record evidence in a labelled table, diagram, or sentence notes.
5. Write a conclusion that answers the question without adding unsupported guesses.
6. Add one safety rule and one limitation of the evidence.

Extension: suggest one improved investigation for a future lesson.

Glossary

Use this glossary to revise important words in Sayansi.

- Living Things Plants And Animals: Explain this term using an example from the book, your school, or your community.
- Water Air And Weather: Explain this term using an example from the book, your school, or your community.
- Food Health And Hygiene: Explain this term using an example from the book, your school, or your community.
- Materials And Their Uses: Explain this term using an example from the book, your school, or your community.
- Forces Movement And Simple Tools: Explain this term using an example from the book, your school, or your community.
- Light Sound And Heat Around Us: Explain this term using an example from the book, your school, or your community.
- Environment Care And Safety: Explain this term using an example from the book, your school, or your community.
- Primary Science Project: Explain this term using an example from the book, your school, or your community.

Add five more words from your lessons and write your own meanings.

Answer And Teacher Notes

A strong Sayansi answer states the idea, uses correct scientific vocabulary, records observations or data, includes units or labels where needed, and writes a conclusion that follows from evidence. Safety rules always come first. If evidence is missing, repeat the observation safely or explain what evidence would be needed.

Final Project

Create a final project for Sayansi that shows what you can now do independently.

Your project must include:

1. A clear title.
2. A real-life problem, question, text, performance, investigation, design, or worked task.
3. Evidence from at least three lessons in this book.
4. A drawing, table, map, chart, model, dialogue, paragraph, performance plan, practical product, or worked solution.
5. A correction note showing one improvement after feedback.
6. A short reflection explaining what you learned, what was difficult, and what you will practise next.

Presentation check: keep the work neat, label important parts, use correct vocabulary, and be ready to explain your choices to a classmate or teacher.